Unnamed_Unit

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738070006237-Chapter 5 - Unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.

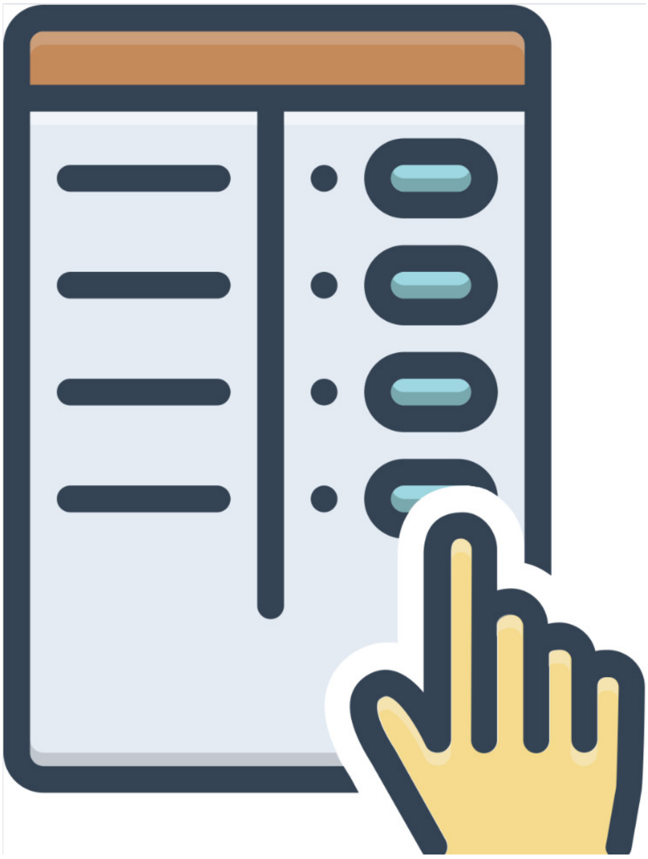
UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

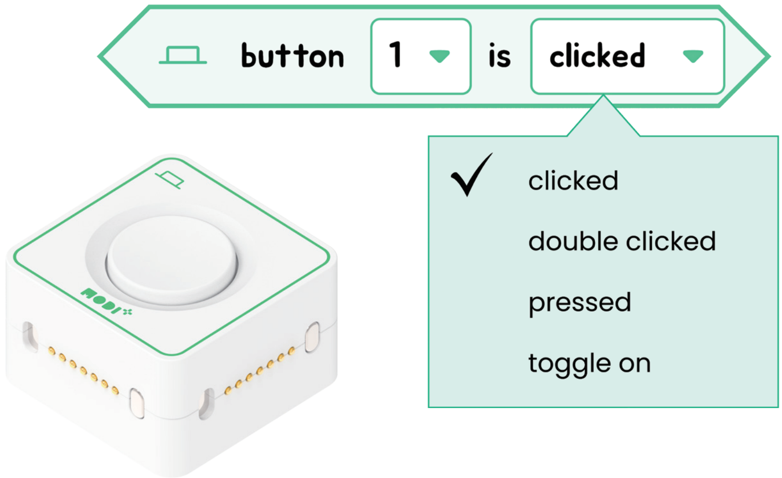
JOHN'S DIGITAL BALLOT: REVOLUTIONIZING SCHOOL ELECTIONS

John had always been fascinated by how decisions were made in his school. From choosing the theme of the winter dance to electing student council officers, it all boiled down to voting. However, last year's student council election had been chaotic. Paper ballots were misplaced, and tallying the votes took hours, leading to disputes about the accuracy of the results. This experience stuck with John, and as he learned more about democracy and voting systems in his social studies class, an idea began to form.

Inspired by a class project on technological solutions for everyday problems, John decided to build a digital voting machine for his school's upcoming elections. He knew that a reliable and transparent voting process was crucial for fair outcomes, and with his new knowledge from the "Humanity & Social Science" unit, he was ready to make a difference. Using a MODI kit, John planned to create a simple yet effective machine that could handle votes electronically, reducing errors and speeding up the counting process. The machine would use a joystick for selections, and each direction would represent a different candidate. His aim was to demonstrate the machine at the next student council meeting, hoping to show how technology could enhance the democratic process right at their school.



ABOUT BUTTON MODULE



1. Click

Detects if Button is clicked or not.
Range: True(Click once) : 100 / False : 0

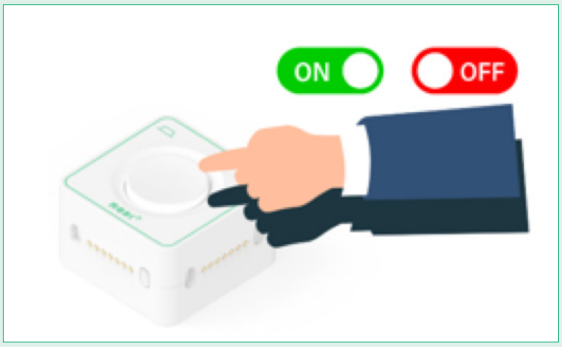
2. Double click

Detects if Button is double - clicked or not.
Range: True(Click twice) : 100 / False : 0

3. Press

Show the press status of Button.
Range: True(Pressed) : 100 / False : 0

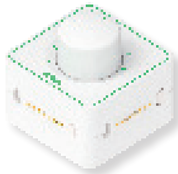
4. Toggle



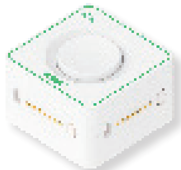
Changes on / off when Button is clicked.
Range: True(On) : 100 / False : 0

Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?

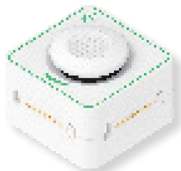


Dial

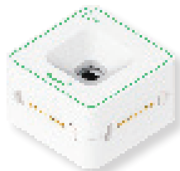


Button

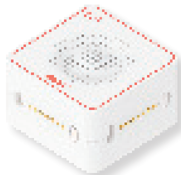
Environment



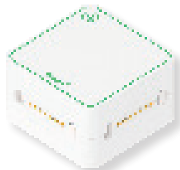
Joystick



ToF

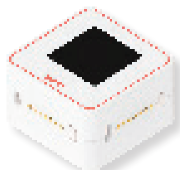


Speaker



IMU

LED

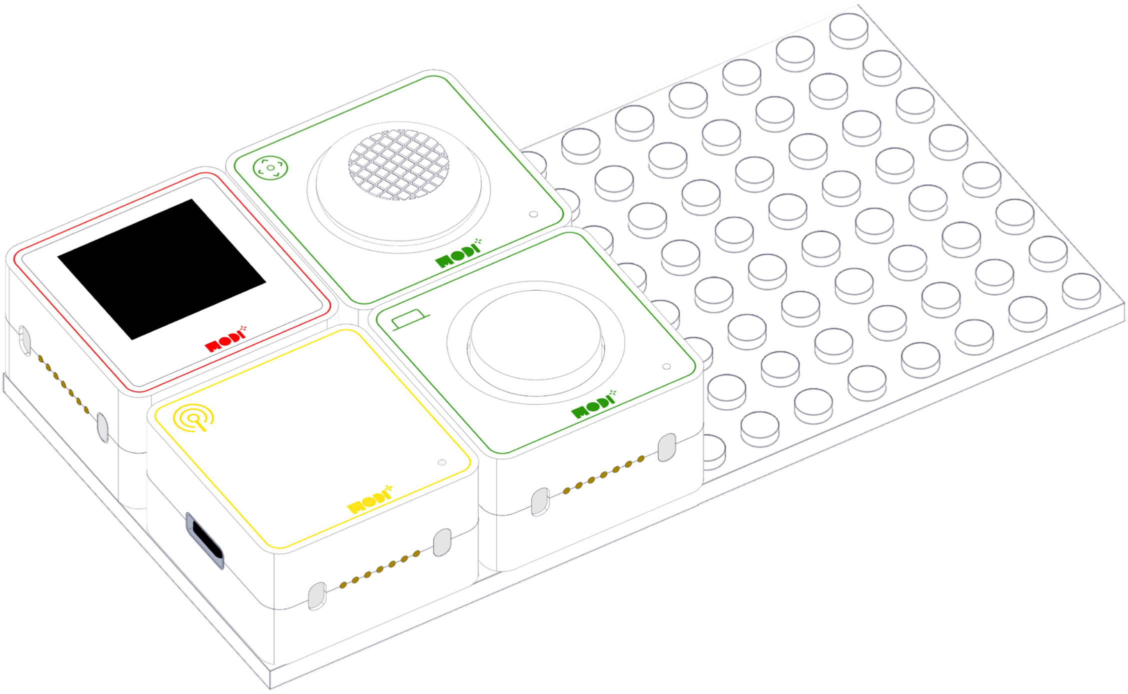


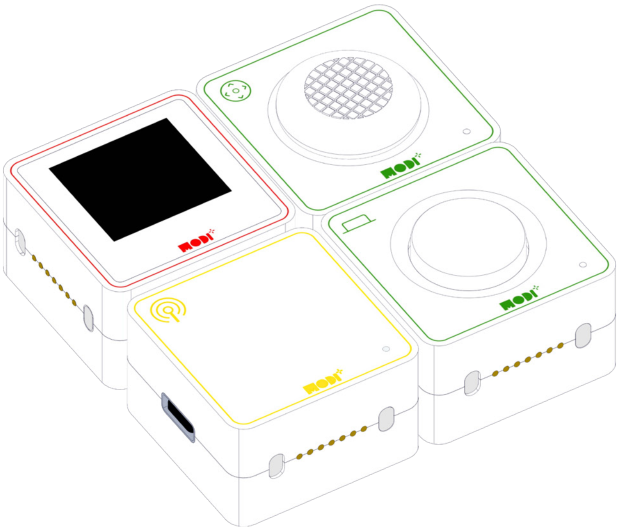
Display

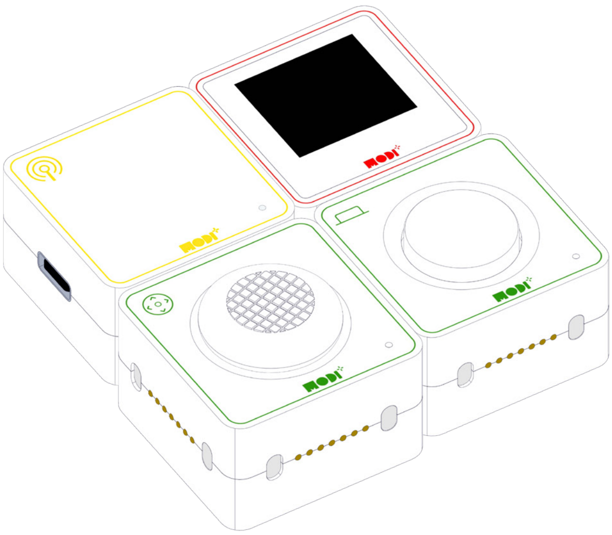
Motor A

Motor B

HOW DOES THE CREATION LOOK LIKE?

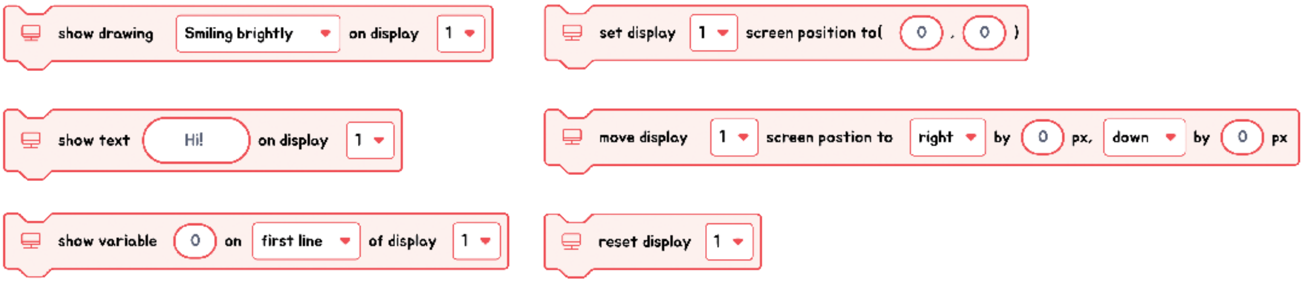






Module Functionality

Let's check the details of the Display module below for this project.

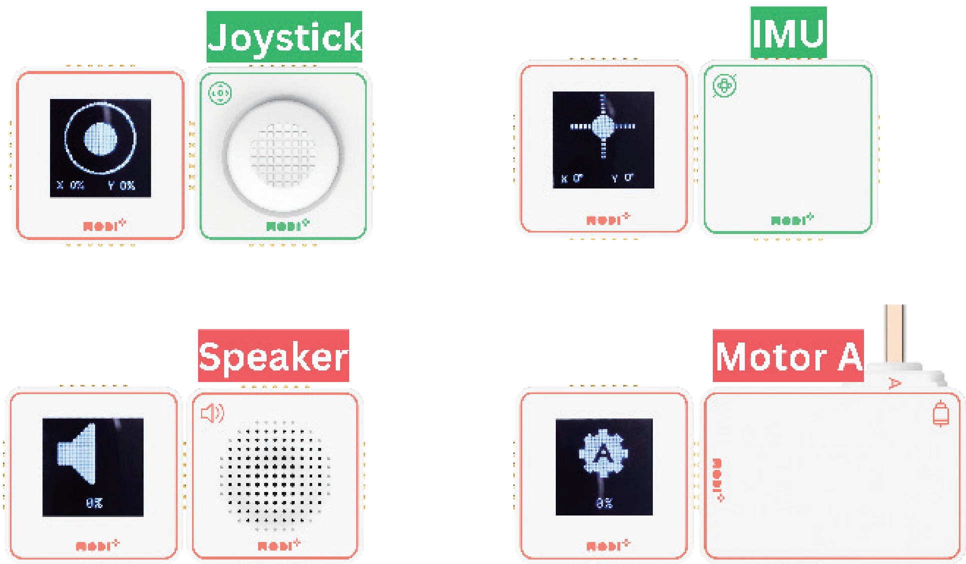


The Display module is like a tiny screen for your MODI projects. It uses 6 different blocks in the Code Editor to show pictures or information from other MODI modules nearby. So, when other modules talk to the Display, it can show you what they're saying or doing with images or data right on its screen!

AUTOMATIC DATA DISPLAY WITH MODI MODULES

The Display module is like the heart of a conversation among MODI modules. Imagine it as a screen that lights up with information whenever other modules, like the Button or the Light Sensor, have something to "say." When any input or output module is connected to it, the Display module automatically picks up their signals and shows what they're doing on its screen.

INTERACTION BETWEEN DISPLAY AND OTHER MODULES



So, if you press a Button module or cover a Light Sensor, the Display will show these actions without needing any special programming. It's like having a translator that instantly shows you what's happening in the MODI module language!